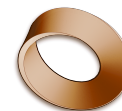


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.9 – X-rays & Gamma Rays

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can explain how X-rays are used to create images of bones inside the body.
- b) I can define **ionising radiation** and identify which types of waves on the electromagnetic spectrum have this property.
- c) I can evaluate the risks and benefits of using ionising radiation in medicine and industry.

Theory

The electromagnetic spectrum continues beyond ultraviolet light into two very high-energy regions: **X-rays** and **gamma rays**. X-rays have shorter wavelengths and higher frequencies than ultraviolet light. Because they carry so much energy, they can pass through materials that visible light cannot. For example, X-rays easily penetrate soft tissue like skin and muscle, but they are absorbed by denser materials like bone. This difference in absorption creates a **shadow image** on a detector or film, allowing doctors to see broken bones or dental issues without surgery.

Gamma rays have the shortest wavelengths and the highest frequencies of all electromagnetic waves. They are produced during nuclear reactions and radioactive decay. Both X-rays and gamma rays, along with some ultraviolet light, are classified as **ionising radiation**. This means they carry enough energy to knock electrons off atoms, turning them into charged particles called ions. This process can damage living cells and DNA, which is why exposure must be carefully controlled.

Despite their dangers, ionising radiation has vital uses in society. In medicine, X-rays are used for diagnostic imaging, while gamma rays are used to **sterilise** medical equipment by killing bacteria and viruses. They are also used in radiotherapy to target and destroy cancer cells. However, because they can cause harm, strict safety protocols are required when handling these waves to protect both patients and workers.

Word bank: absorbed · ionising · sterilise · shadow · DNA

Activity

Investigating Penetration Power

You will need: a torch (flashlight), a sheet of paper, a piece of cardboard, and a thick book.

- Turn on the torch and shine it at a wall in a dark room. Note how bright the spot is.
- Place the sheet of paper between the torch and the wall. Observe the light. Does it pass through?
- Replace the paper with the cardboard. Observe the light again. Is the spot dimmer?
- Finally, place the thick book in the path of the light. What happens to the light?

Material	Light Passes Through? (Yes/No)	Relative Density (Low/Medium/High)
Paper	_____	Low
Cardboard	_____	Medium
Book	_____	High

Think about how this relates to X-rays. If visible light is blocked by a book, what does that tell you about the energy required for X-rays to pass through bone?

Questions

1. Explain why bones appear white on an X-ray image while soft tissue appears dark.

2. What is the key difference between ionising and non-ionising radiation?

3. Give one example of how gamma rays are used in medicine and explain why they are suitable for this purpose.

4. **Criterion D (Evaluating):** A hospital wants to use X-rays to diagnose a patient with a broken leg, but the patient is pregnant. Discuss the risks and benefits. Should the procedure go ahead? Justify your answer using scientific evidence.

Extension

Research how gamma rays are used in food irradiation. How does this process help keep food fresh for longer without making it radioactive?
